

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 7. String and its Methods

Lecture: 79. More Inquiry Methods

Section 1: Lecture Summary

This lecture focuses on Python string inquiry methods that return Boolean values (True or False). The main methods covered are `isdigit()`, `isdecimal()`, `isnumeric()`, `isascii()`, and `isalnum()`. These methods help determine the nature of the characters in a string with respect to digits, decimal numbers, numeric characters across languages, ASCII range characters, and alphanumeric characters from various languages. The instructor explains each method's behavior in detail, differentiates closely related methods by examples including Unicode characters (subscripts, superscripts, and numerals from other languages like Devanagari), and encourages hands-on practice in Jupyter Notebook to grasp their practical use. The lecture also touches upon the Unicode and ASCII code ranges briefly, indicating a future detailed topic on them.

Section 2: Key Concepts and Explanations

- Inquiry methods and Boolean results

All methods discussed return `True` or `False` based on the string content.

- `str.isdigit()`

- Returns `True` if all characters in the string are digits.
- Digits include standard numbers (`'0-9'`), Unicode subscript digits (`'₀' to '₉'`), and superscript digits (`'⁰' to '⁹'`).
- Returns `False` if string contains decimals points, minus signs, alphabets, or special characters.
- Example: `'7235'.isdigit() → True`; `'7₂3⁵'.isdigit() → True`; `'71.23'.isdigit() → False`.

- `str.isdecimal()`

- Returns `True` if all characters in the string are decimal characters that can be used to form numbers in decimal systems.
- Supports decimal numerals from various language scripts (e.g., Devanagari numerals).
- Does not accept superscripts, subscripts, fractions, or non-decimal numerals.
- Returns `False` if characters include alphabetic, minus, dot, or special symbols.
- Example: Devanagari "३७५४२" is `True` for `isdecimal()`.
- More strict than `isdigit()` in allowed characters.

- `str.isnumeric()`

- Returns `True` if all characters are numeric characters.
- Broader than `isdigit()` and `isdecimal()`, includes digits, subscripts, superscripts, fractions (like `¾`) and numeric characters from all languages.
- Returns `False` if the string contains punctuation, alphabet, symbols, or signs.
- Example: `¾` (Unicode fraction) returns `True`.

- Differences among `isdigit()`, `isdecimal()`, and `isnumeric()`

- `isdecimal()` only for decimal digits (0-9, including language-dependent decimal digits).
- `isdigit()` includes `isdecimal()` plus digit-like characters (superscripts and subscripts).
- `isnumeric()` includes `isdigit()` plus numeric characters (like fractions) in Unicode.

- `str.isascii()`

- Returns `True` if all characters in the string are ASCII characters, i.e., Unicode code points 0 to 127.
- Only English language characters, digits, and special ASCII symbols qualify.
- Any character outside ASCII range returns `False`.

- `str.isalnum()`

- Returns `True` if all characters are alphanumeric (letters and numbers) without spaces or special characters.
- Letters and digits can be from any language or script.
- Special characters and non-printable characters cause `False`.

- **Unicode and ASCII basics**

- Unicode range for ASCII characters is U+0000 to U+007F (decimal 0 to 127).
- Unicode allows representing digits and letters in various languages/scripts, impacting the results of these inquiry methods.

Section 3: Example Code and Use Cases

Check if string contains only digits, including subscript and superscript digits

String with decimal point returns False for `isdigit()`

Check if string is decimal (Devanagari numerals example)

Check if numeric (includes fractions)

Check ASCII characters

Check alphanumeric with English and other languages

```
s = '7\u2082\u2075' # 7 subscript 2 3 superscript 5
print(s)           # Output: 7235
print(s.isdigit()) # True

s2 = '71.23'
print(s2.isdigit()) # False

s3 = '\u0969\u096a\u096b' # Devanagari digits 3, 4, 5
print(s3)                 # Prints those digits
print(s3.isdecimal())     # True

s4 = '\u00BC' # Unicode for ¼ fraction
```

```
print(s4.isnumeric())      # True

s5 = 'Hello123'
print(s5.isascii())        # True

s6 = 'नमस्ते123' # Hindi text with digits
print(s6.isascii())        # False

s7 = 'abc123सदैव'
print(s7.isalnum())        # True

s8 = 'abc123!'
print(s8.isalnum())        # False due to '!'
```

Each code block demonstrates checking the string with respective inquiry methods to understand what counts as true for digits, decimals, numerics, ASCII characters, and alphanumeric strings.

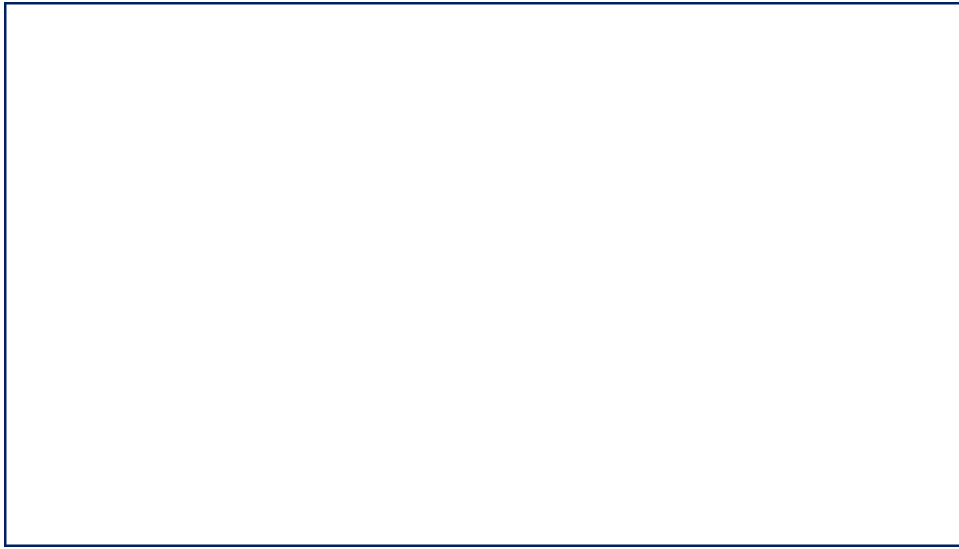
Section 4: Key Takeaways

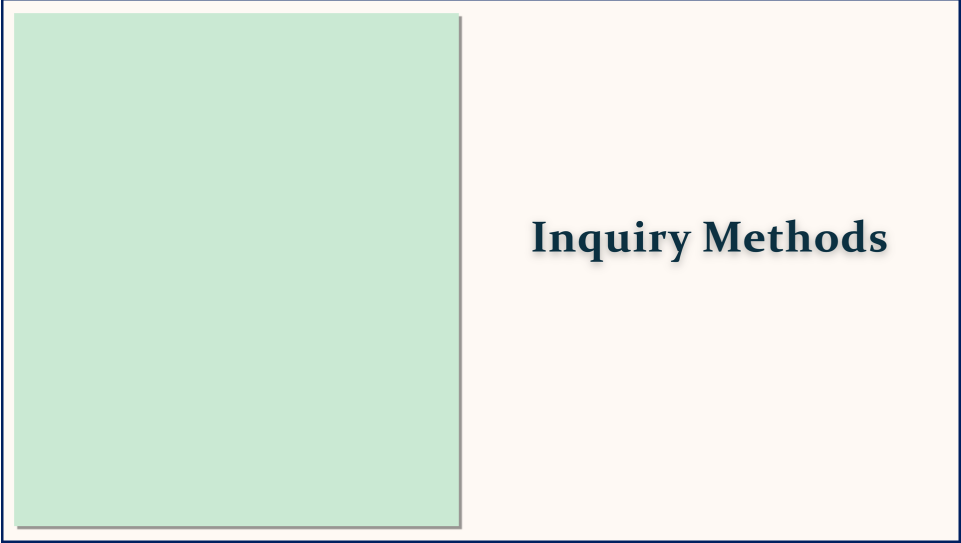
- Inquiry string methods return Boolean values indicating if strings match specific character criteria.
- `isdigit()` returns `True` for standard digits plus Unicode subscript/superscript digits only — no decimals or symbols.
- `isdecimal()` is stricter, only for decimal digits including those in other languages, excluding superscripts/subscripts and fractions.
- `isnumeric()` is the most inclusive for numeric values, including digits, superscripts, subscripts, fractions, and numeric symbols from all scripts.
- `isascii()` checks if all characters belong to ASCII (0-127 code points), mainly English letters, digits, and common symbols.
- `isalnum()` allows alphanumeric characters (letters and digits) from any language but no special characters or spaces.
- Unicode plays a critical role in determining how these methods interpret string content, as different scripts and symbols are classified differently.

- Understanding these distinctions helps in proper input validation and processing when dealing with strings containing multilingual numerals or symbols.
- Practical experimentation using Jupyter Notebook, especially using Unicode codes, deepens understanding of these methods.

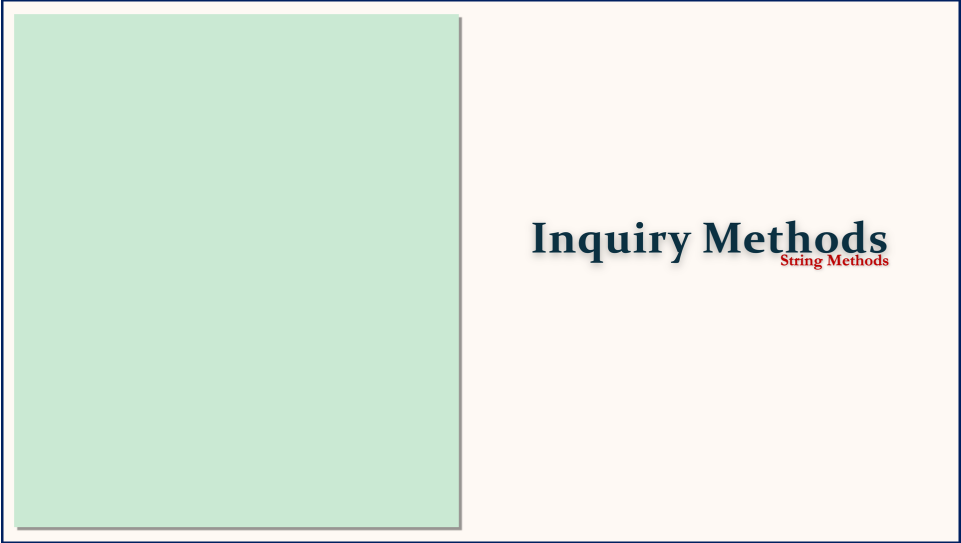
Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.





Inquiry Methods



Inquiry Methods

String Methods

String Methods

String Methods

True / False

Inquiry Methods
String Methods

isdigit ()

Inquiry Methods
String Methods

isdigit ()

isdecimal ()

Inquiry Methods
String Methods

isdigit ()

isdecimal ()

isnumeric ()

Inquiry Methods
String Methods

isdigit ()

isdecimal ()

isnumeric ()

isascii ()

Inquiry Methods

String Methods

isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()

Inquiry Methods

String Methods

isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()

Inquiry Methods

String Methods

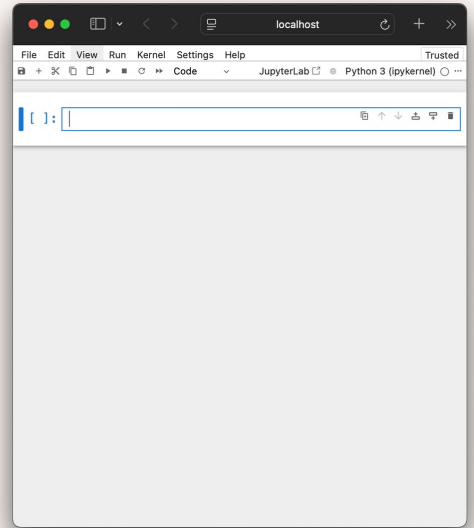
`isdigit ( )`

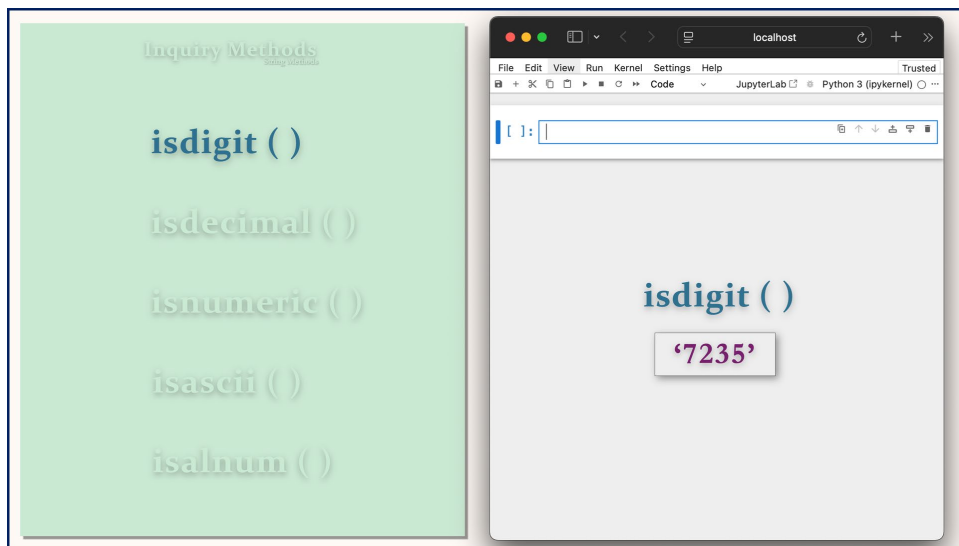
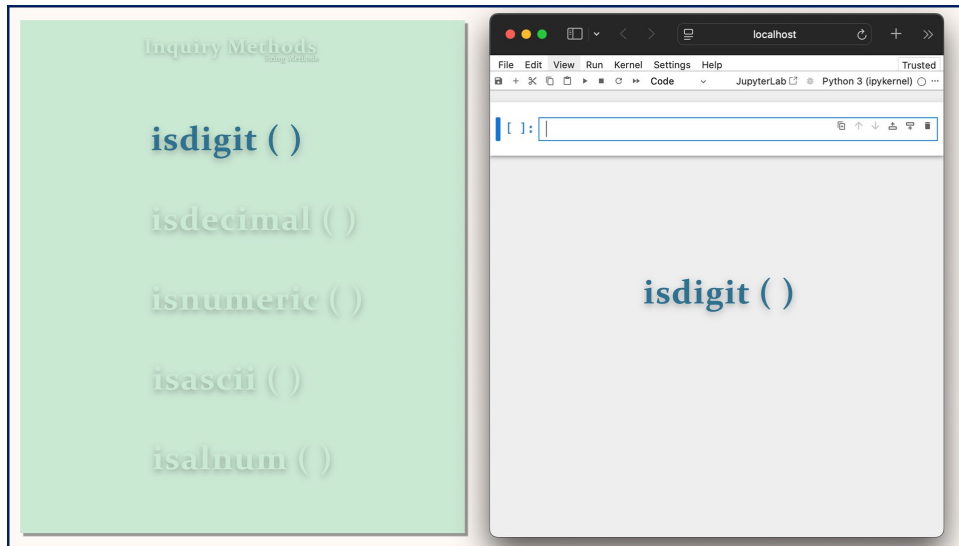
`isdecimal ( )`

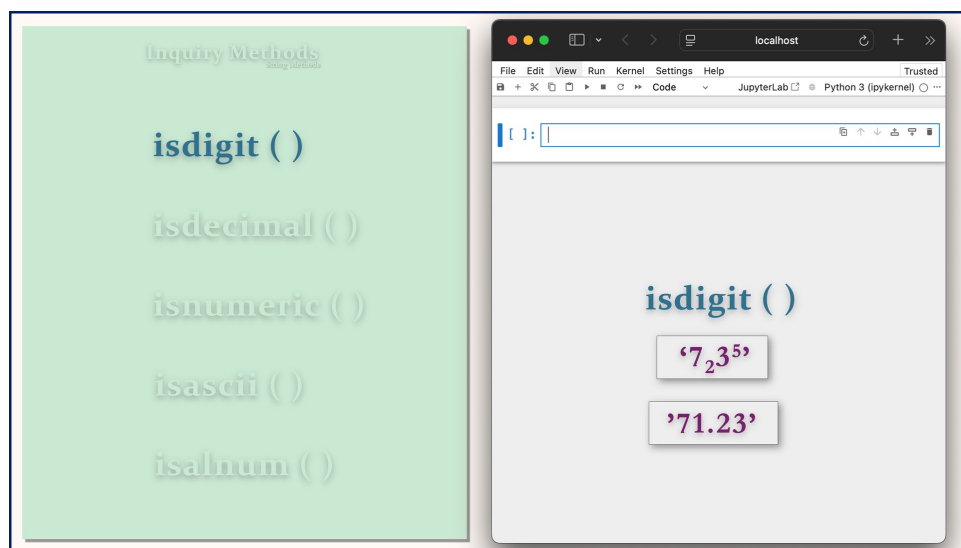
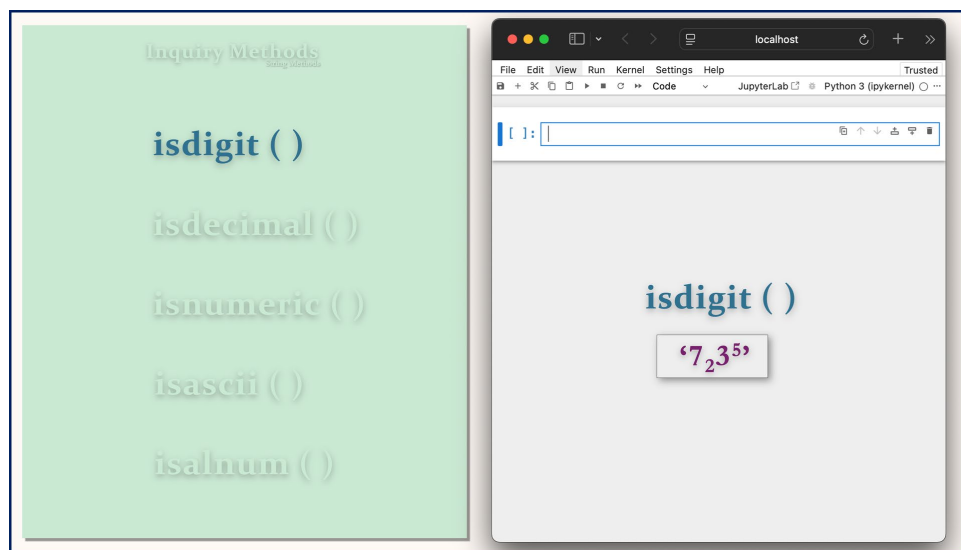
`isnumeric ( )`

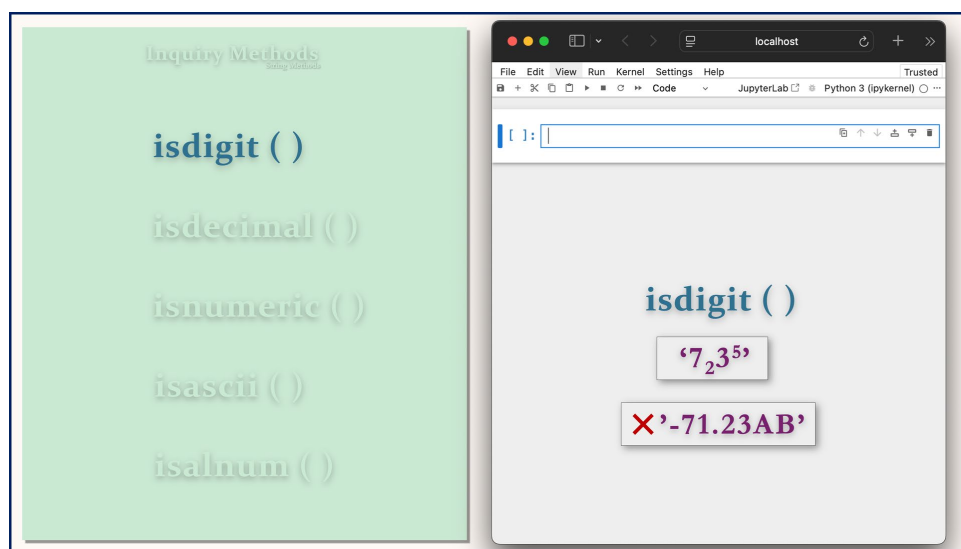
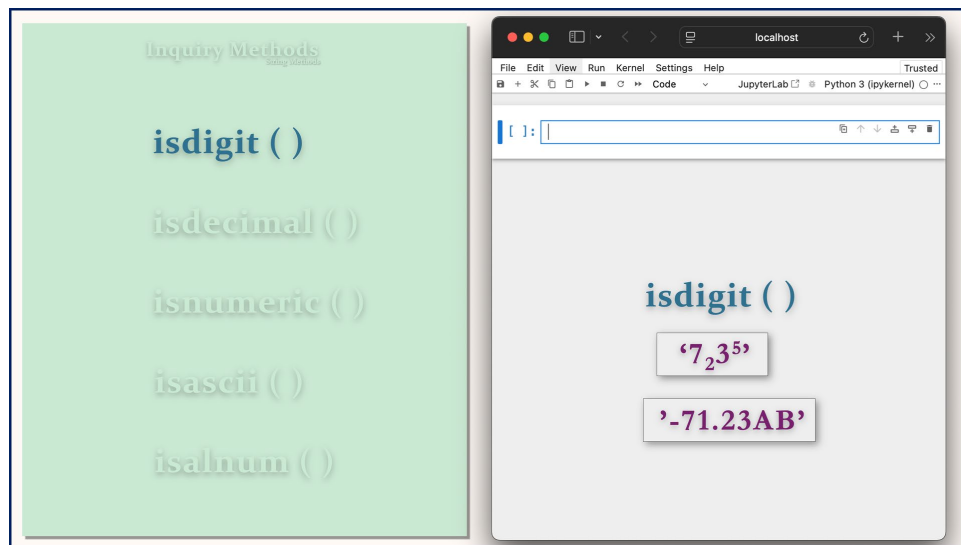
`isascii ( )`

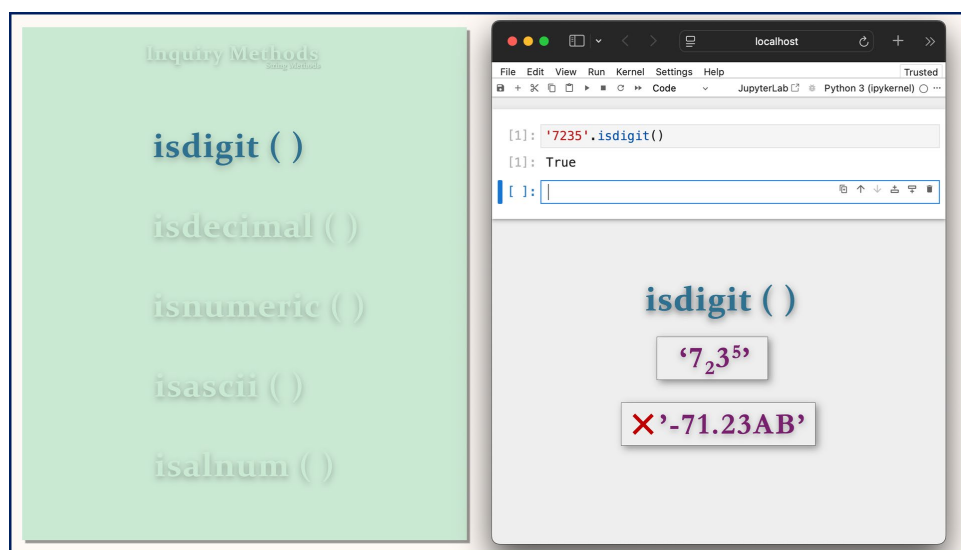
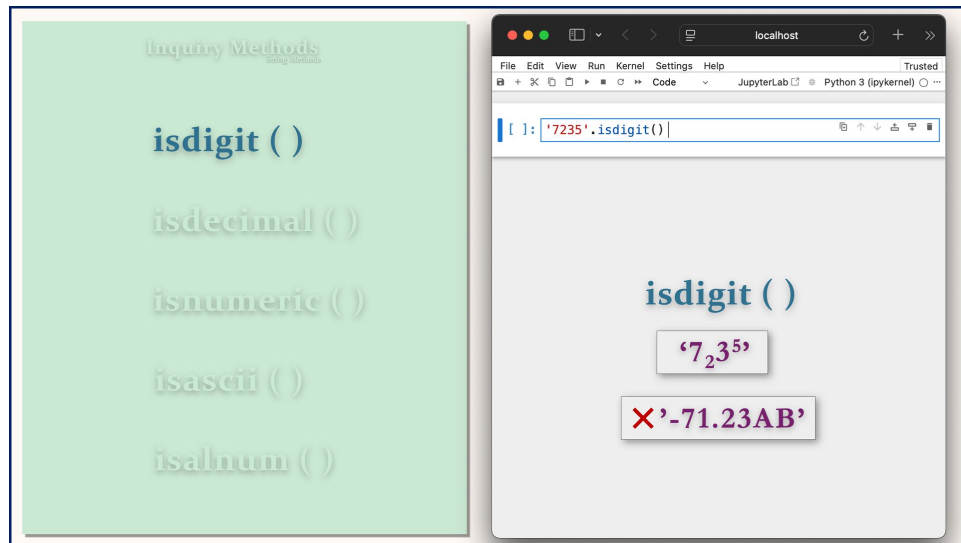
`isalnum ( )`











Inquiry Methods

isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()

localhost

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JupyterLab Python 3 (ipykernel)

[]: s = '7\u2082\u2083'

isdigit ()

'7₂3⁵'

✗'-71.23AB'

Inquiry Methods

isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()

localhost

File Edit View Run Kernel Settings Help

JupyterLab Python 3 (ipykernel)

[]: s = '7\u2082\u2083\u2085\u2087'

isdigit ()

'7₂3⁵'

✗'-71.23AB'

Inquiry Methods

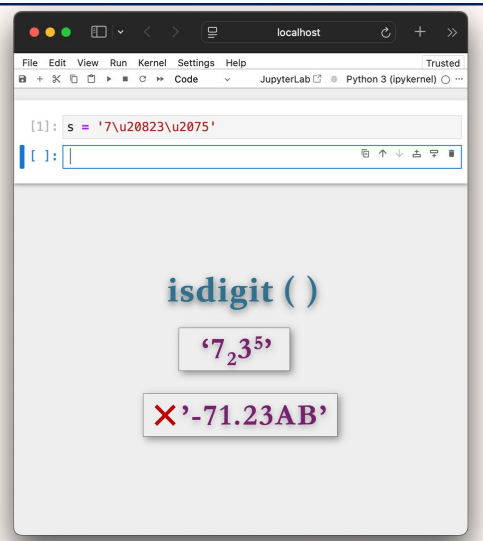
isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()



Inquiry Methods

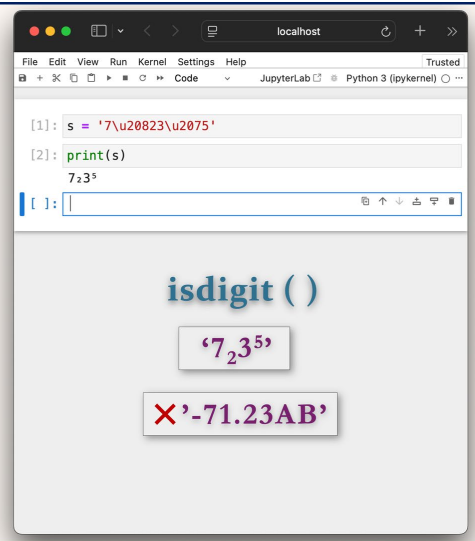
isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()



```
[1]: s = '7\u20823\u2075'
[2]: print(s)
7\u20823\u2075
```

isdigit ()

'7₂3⁵'

✗'-71.23AB'

Inquiry Methods

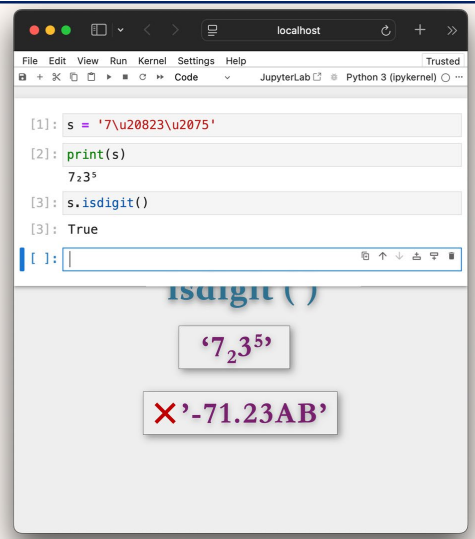
isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()

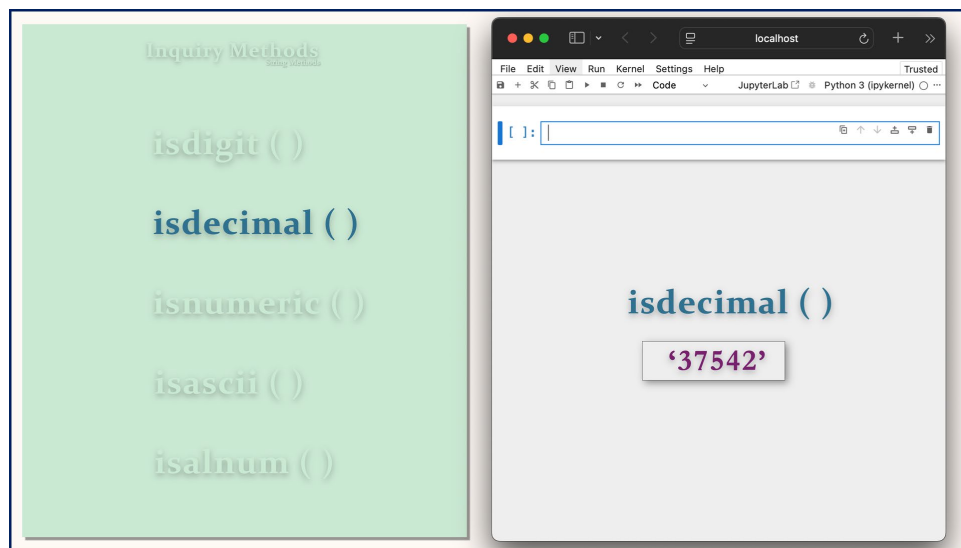
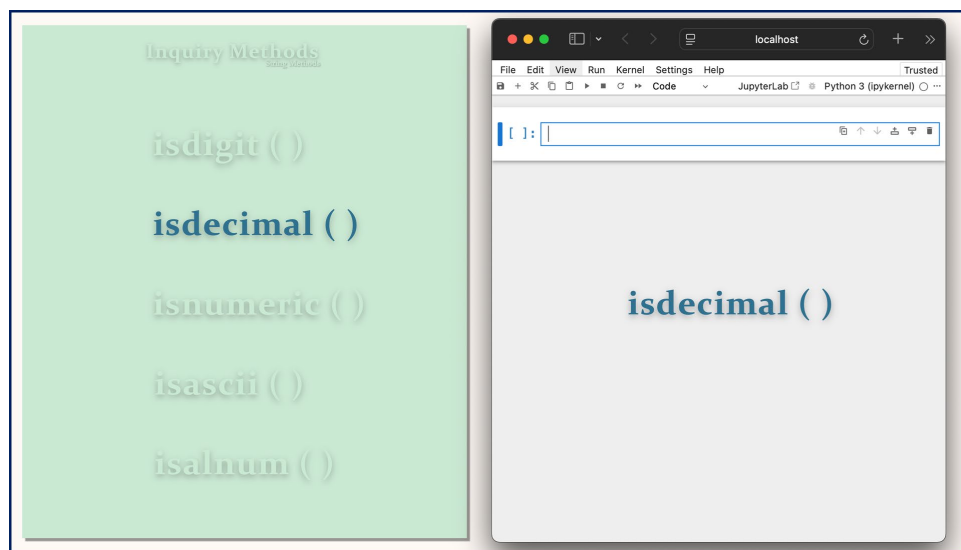


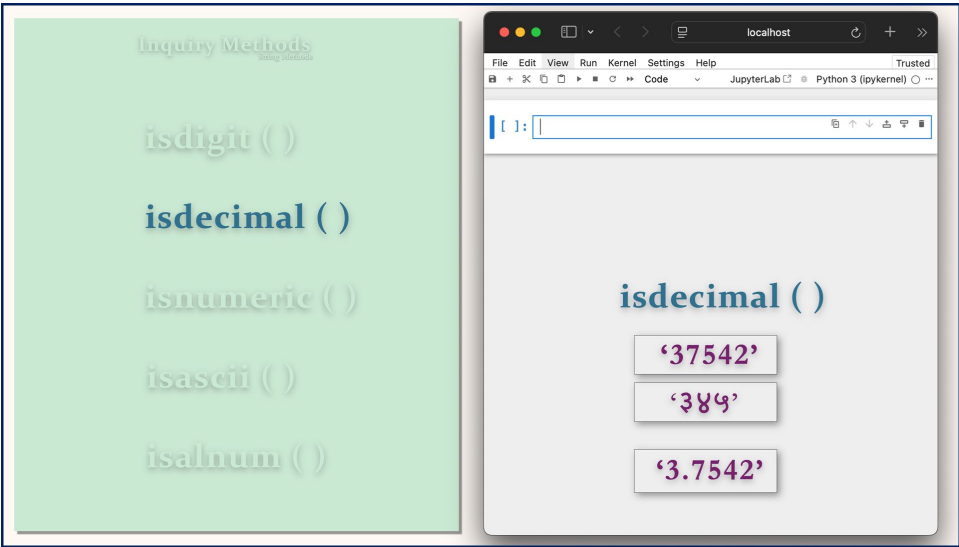
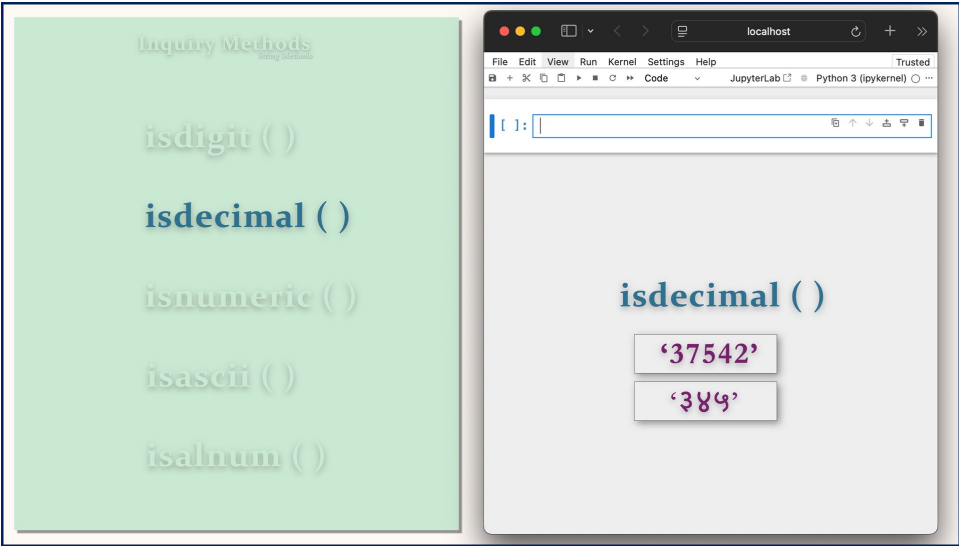
```
[1]: s = '7\u20823\u2075'
[2]: print(s)
7\u20823\u2075
[3]: s.isdigit()
True
```

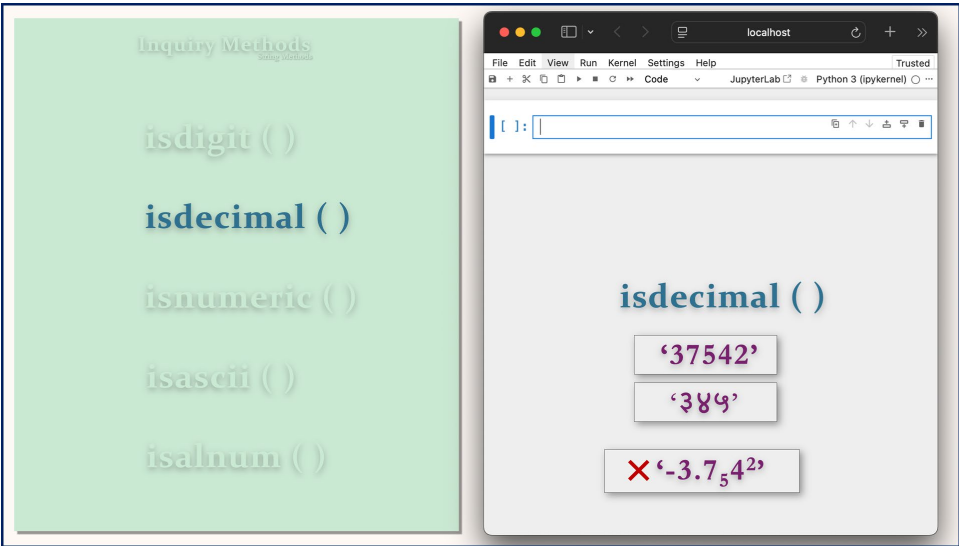
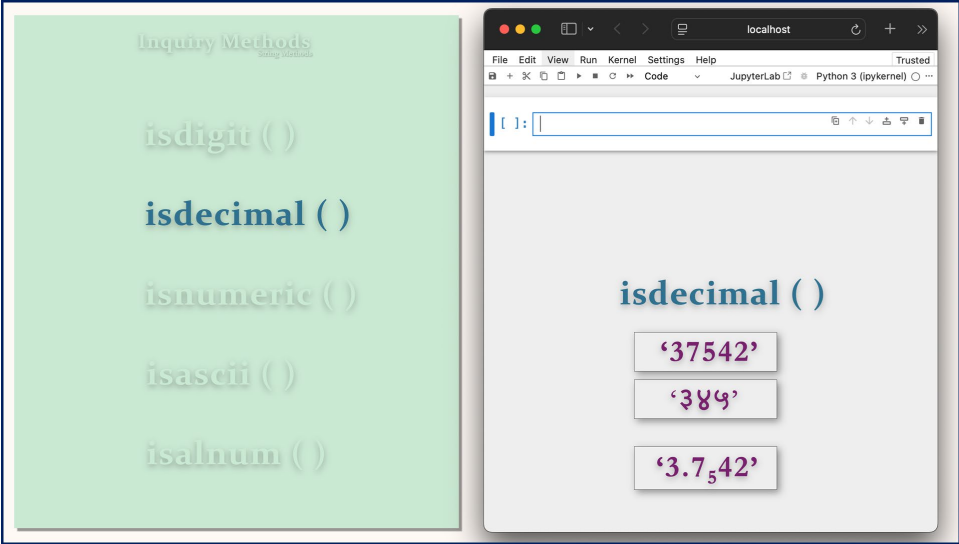
isdigit ()

'7₂3⁵'

✗'-71.23AB'







isdecimal ()

isdecimal ()

‘37542’

‘384’

✗ $'-3.7_5 4^2'$

isdecimal ()

isdecimal ()

‘37542’

‘384’

✗ $'-3.7_5 4^2'$

Inquiry Methods

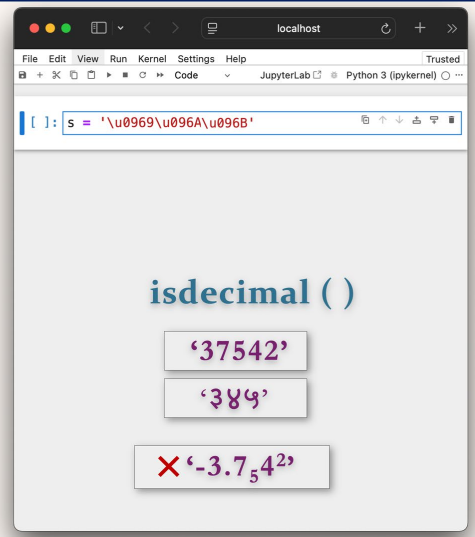
`isdigit ( )`

`isdecimal ( )`

`isnumeric ( )`

`isascii ( )`

`isalnum ( )`



Inquiry Methods
song shiduo

`isdigit ( )`

`isdecimal ( )`

`isnumeric ( )`

`isascii ( )`

`isalnum ( )`

localhost

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JupyterLab Python 3 (pykernel)

```
[1]: s = '\u0969\u096A\u096B'
```

```
[ ]:
```

`isdecimal ( )`

`'37542'`

`'३४५'`

✗ `'-3.7542'`

Inquiry Methods
song shiduo

`isdigit ( )`

`isdecimal ( )`

`isnumeric ( )`

`isascii ( )`

`isalnum ( )`

localhost

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (pykernel)

```
[1]: s = '\u0969\u096A\u096B'
```

```
[2]: print(s)
```

```
3४५
```

```
[ ]:
```

`isdecimal ( )`

`'37542'`

`'३४५'`

✗ `'-3.7542'`

Inquiry Methods

isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()

```
[1]: s = '\u0969\u096A\u096B'
```

```
[2]: print(s)
```

```
३४५
```

```
[3]: s.isdecimal()
```

```
[3]: True
```

isdecimal ()

'37542'

'३४५'

✗'-3.7,4²'

Inquiry Methods

isdigit ()

isdecimal ()

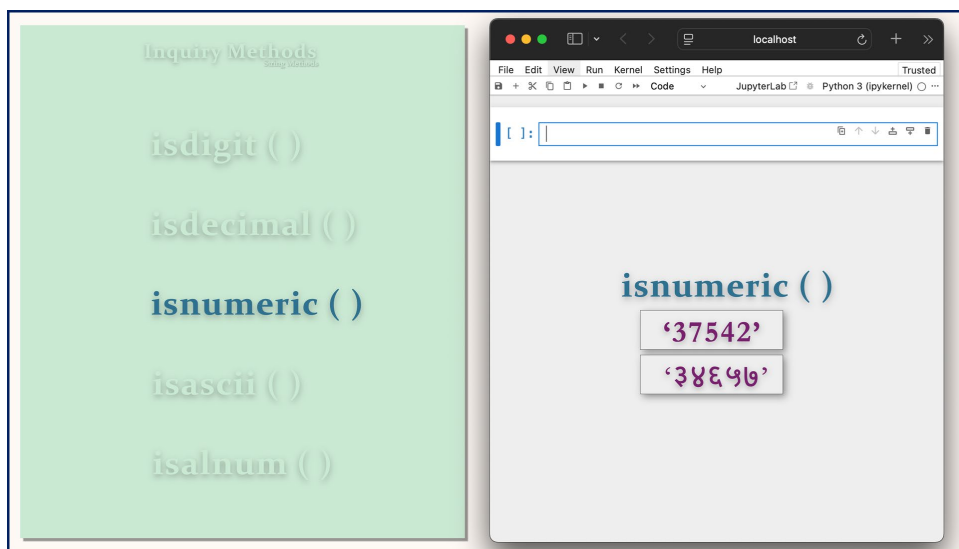
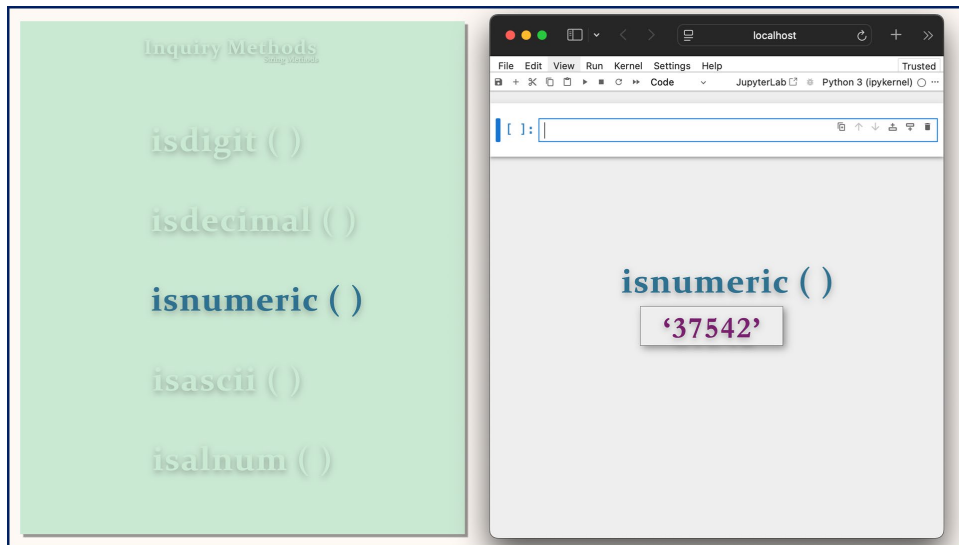
isnumeric ()

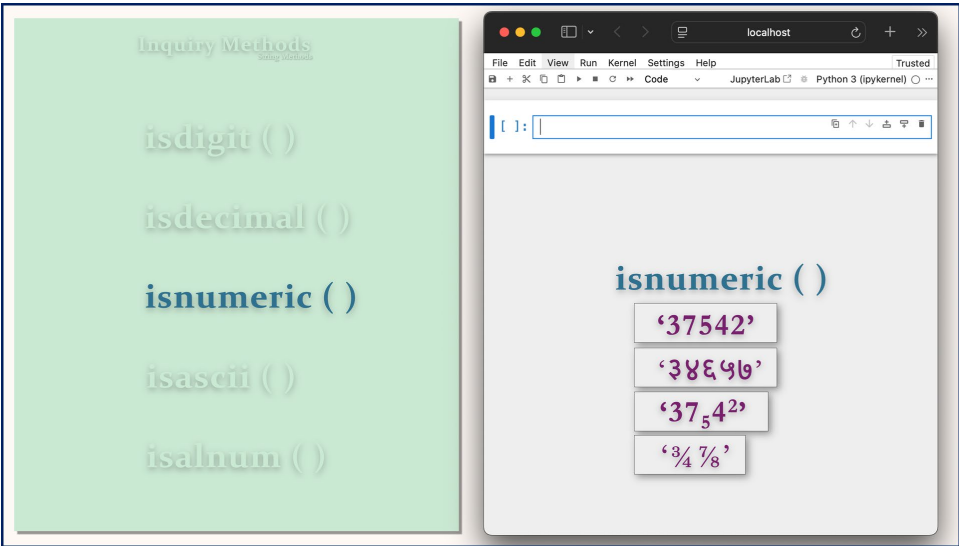
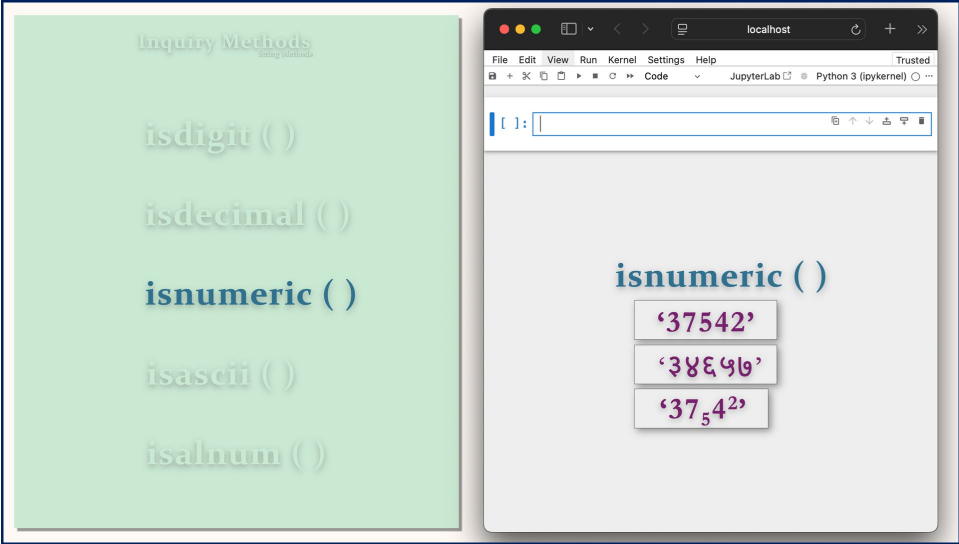
isascii ()

isalnum ()

```
[ ]:
```

isnumeric ()





isnumeric ()

isnumeric ()

‘37542’

‘३४६५७’

'37₅4²'

$\frac{3}{4} \frac{7}{8}$

‘3.75’

isnumeric ()

isnumeric ()

‘37542’

‘ဒုက္ခဒုခ’

‘37₅4²’

$\frac{3}{4} \frac{7}{8}$

‘3.75ab’

Inquiry Methods

`isdigit ( )`

`isdecimal ( )`

`isnumeric ( )`

`isascii ( )`

`isalnum ( )`

`isnumeric ( )`

`'37542'`

`'38&99'`

`'37542'`

`'3/4 7/8'`

`'-3.75ab'`

Inquiry Methods
song shiduo

isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()

localhost

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JupyterLab Python 3 (ipykernel)

[]:

isnumeric ()

'37542'

'૩૪૬૫૭'

'37₅4²'

'¾ ⅞'

✗ '-3.75ab'

Inquiry Methods
song shiduo

isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()

localhost

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (ipykernel)

[]: s = '\u00BE'

isnumeric ()

'37542'

'૩૪૬૫૭'

'37₅4²'

'¾ ⅞'

✗ '-3.75ab'

Inquiry Methods

isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()

```
[ ]: s = '\u00BE\u215E'
```

isnumeric ()

'37542'

'٣٨٤٩٦'

'37₅4²'

'¾ ⅞'

✗ '-3.75ab'

Inquiry Methods

isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()

```
[1]: s = '\u00BE\u215E'
```

```
[ ]:
```

isnumeric ()

'37542'

'٣٨٤٩٦'

'37₅4²'

'¾ ⅞'

✗ '-3.75ab'

Inquiry Methods

isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()

```
[1]: s = '\u00BE\u215E'
```

```
[2]: print(s)
```

```
¾
```

```
[ ]:
```

isnumeric ()

'37542'

'¾¾¾'

'37₅4²'

'¾ ¾'

✗ '-3.75ab'

Inquiry Methods

isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()

```
[1]: s = '\u00BE\u215E'
```

```
[2]: print(s)
```

```
¾
```

```
[3]: s.isnumeric()
```

```
[3]: True
```

```
[ ]:
```

isnumeric ()

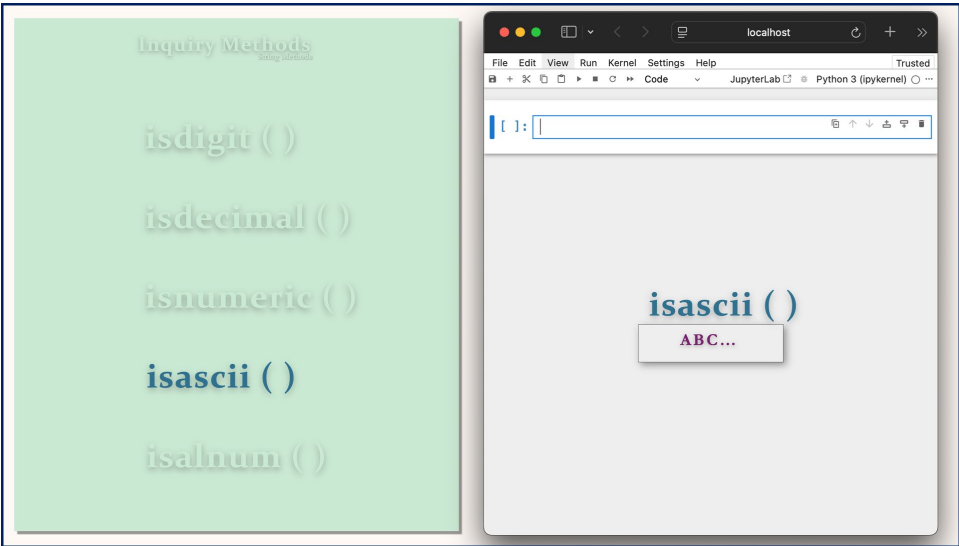
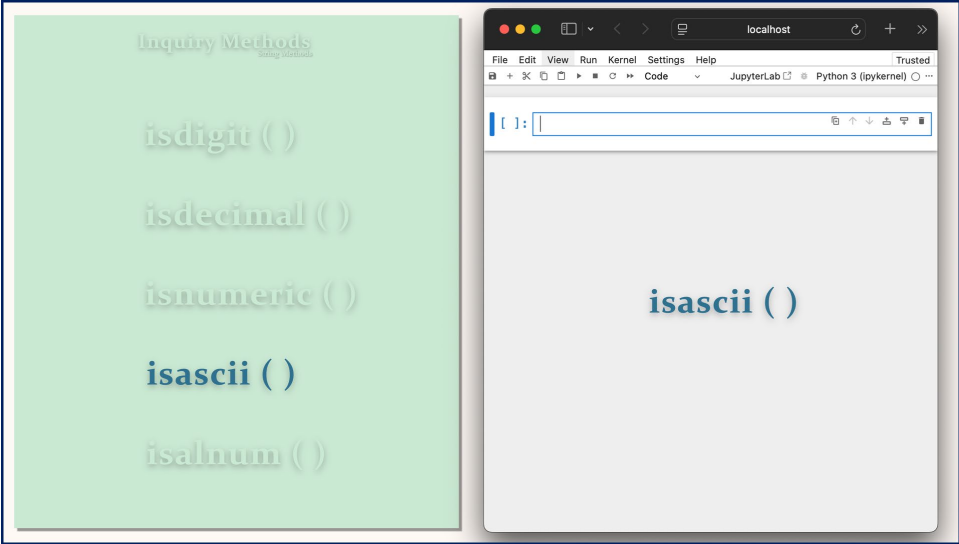
'37542'

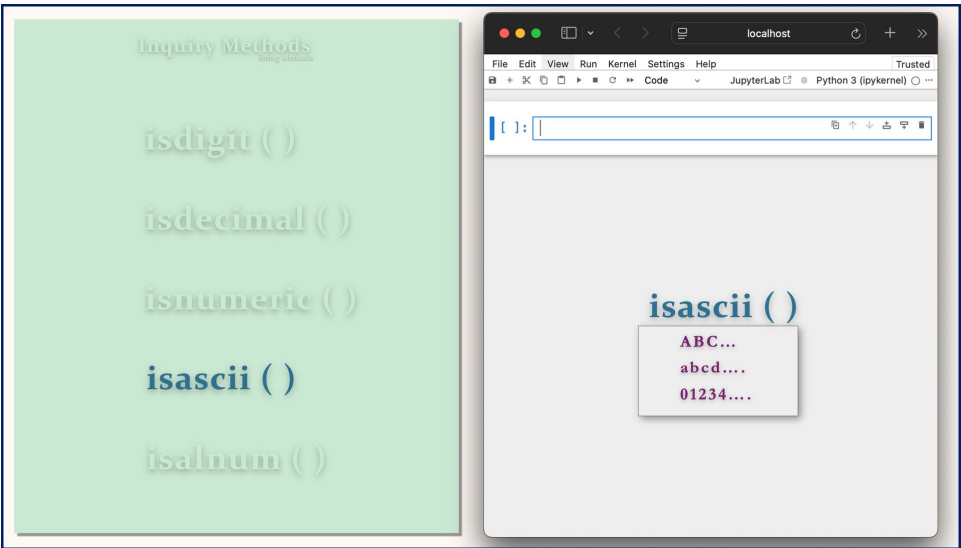
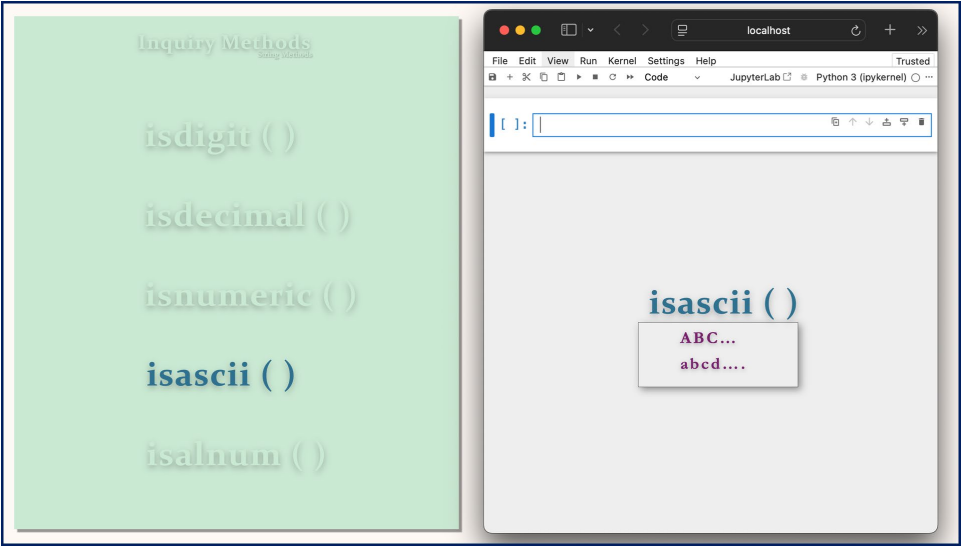
'¾¾¾'

'37₅4²'

'¾ ¾'

✗ '-3.75ab'





Inquiry Methods

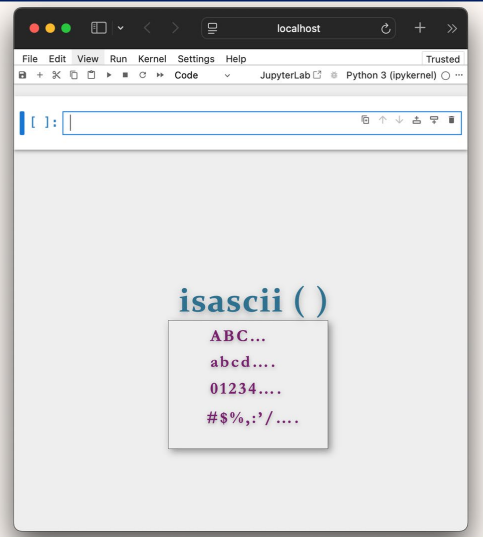
isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()





Inquiry Methods

isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()

localhost

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JupyterLab Python 3 (ipykernel)

[]:

isascii ()

ABC...
abcd....
01234....
#\$%,;' /
\n\r\f\v....

0 - 127
\x00 - \x7f

Inquiry Methods

isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()

localhost

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (ipykernel)

[]:

isascii ()

ABC...
abcd....
01234....
#\$%,;' /
\n\r\f\v....

0 - 127
\x00 - \x7f
\u0000 - \u007f

Inquiry Methods

isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()

localhost

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (ipykernel)

[]:

isascii ()

ABC...
abcd....
01234....
#\$%&,' /
\n\r\f\v....

0 - 127
\x00 - \x7f
\u0000 - \u007f

‘38&9b’

Inquiry Methods

isdigit ()

isdecimal ()

isnumeric ()

isascii ()

isalnum ()

localhost

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (ipykernel)

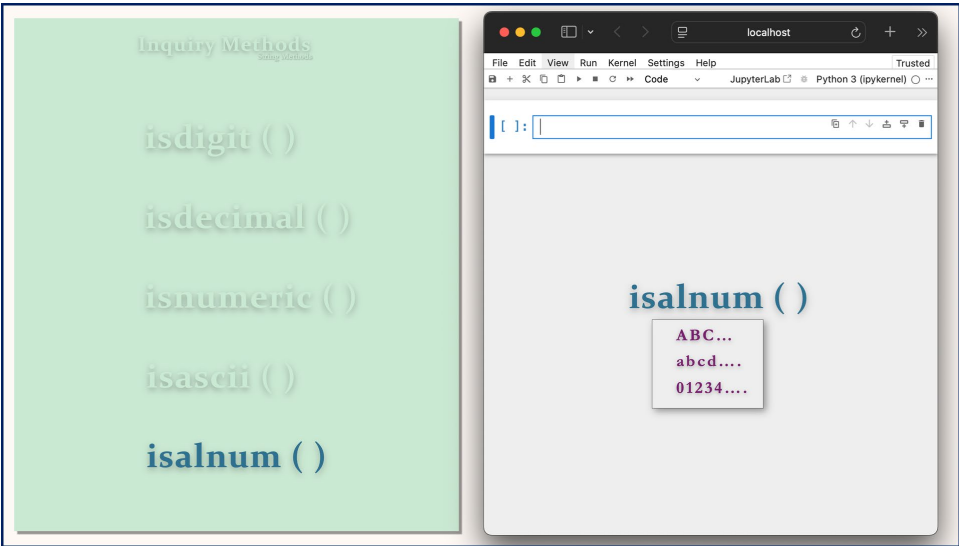
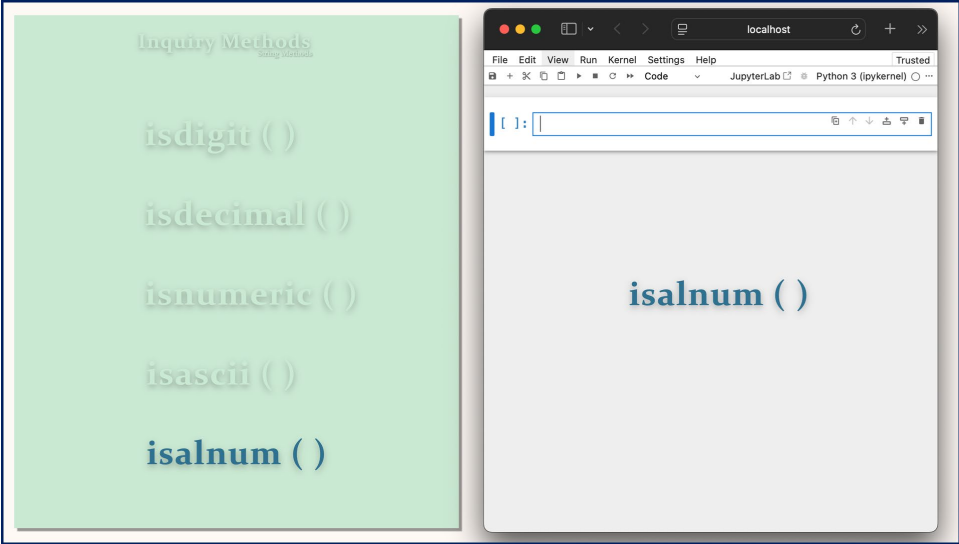
[]:

isascii ()

ABC...
abcd....
01234....
#\$%&,' /
\n\r\f\v....

0 - 127
\x00 - \x7f
\u0000 - \u007f

✗‘38&9b’



Inquiry Methods

`isdigit ( )`

`isdecimal ( )`

`isnumeric ( )`

`isascii ( )`

`isalnum ( )`

